

A Spatial Alignment Framework Using Geolocation Cues for Roadside Multi-View Multi-Sensor Fusion

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Abstract—Multi-modal sensor fusion plays a vital role in achieving high-quality roadside perception for intelligent traffic monitoring. Unlike on-board sensors in autonomous driving, roadside sensors present heightened calibration complexity, posing a challenge to spatial alignment for data fusion. And existing spatial alignment methods typically focus on one-to-one alignment between cameras and radar sensors and require precise calibration. However, when applied to large-scale roadside monitoring networks, these methods can be difficult to implement and may be vulnerable to environmental influences. In this paper, we present a spatial alignment framework that utilizes geolocation cues to enable multi-view alignment across distributed multi-sensor systems. In this framework, a deep learning-based camera calibration model combined with angle and distance estimation is used for monocular geolocation estimation. A camera parameter approaching method is then used to search for pseudo camera parameters that can tolerate inevitable calibration errors in practice. Finally, the geolocation information is then used for data association between Light Detection and Ranging (LiDAR) and cameras. The framework has been conducted and tested at several intersections in Hangzhou. Experimental results show that the framework can achieve geolocation estimation errors of less than 1.1 m for vehicles traversing the monitored zone, demonstrating the framework's ability to accomplish spatial alignment with a singular execution, and apply it in extensive large-scale roadside sensor fusion scenarios.

I. INTRODUCTION

With their ability to percept from beyond the line of sight, roadside sensors are widely used to obtain high-quality traffic perception data in various intelligent transportation systems, such as advanced driver assistance systems (ADAS) [1], cooperative vehicle-infrastructure systems [2], smart highways [3] and so on.

LiDAR, millimeter-wave (MMV) radar, and cameras are crucial roadside sensors for intelligent traffic monitoring. LiDAR provides precise distance measurements and detailed 3D scene reconstruction but can be expensive and sensitive to environmental conditions. MMV radar, on the other hand, offers cost-effectiveness and weather robustness, but it may lack the fine-grained resolution of LiDAR. Cameras capture rich color, texture, and contextual information, but lack depth information and can be affected by factors such as low-light conditions or occlusions. For these reasons, these sensors are often used together by sensor fusion to overcome their individual limitations and improve roadside perception and

robustness. In recent years, various radar-camera or LiDAR-camera integrated systems have been developed to fulfill the requirements of fusion perception. Despite this, there are still a large number of radars (or LiDARs) and cameras already installed on roadsides. To avoid the waste of replacing these devices, additional methods must be employed to enable fusion perception.

A prerequisite for data fusion of scattered sensors is spatial alignment, which involves projecting objects perceived by multiple sensors onto the same spatial coordinate system. Spatial alignment for camera and radar (or LiDAR) data is done through joint calibration, before joint calibration, it is usually necessary to calibrate the camera first. For instance, studies such as [4], [5] utilized the Zhang's method [6] for camera calibration. According to Du *et al.* [7], camera calibration was achieved by solving the homography matrix of four corresponding points and then a spatio-temporal synchronization optimization model is used to correct the pre-calibration file. Wang *et al.* [8] did not perform camera calibration directly. Instead, they combined a large number of related MMV radar and video object pairs, and used least square fitting or interpolation methods to estimate the spatial transformation relationship function between the two coordinate systems for spatial alignment.

For multi-view scenarios, manual chessboard calibration [6] itself is not accurate, as the calibration results can vary significantly across different individuals. Due to the elevated positions at which cameras are often installed in roadside scenarios, manual calibration becomes impractical, as previous methods primarily relied on simulated data for testing and did not account for this practical issue. The method of solving the homography matrix using four corresponding points [7] also requires manual selection of corresponding points between images and scenes, introducing human errors. Additionally, cameras often experience angle changes due to physical factors, and with the increasing use of Pan-Tilt-Zoom (PTZ) cameras, cameras parameters can change at any time, conducting manual calibration at all times is unrealistic and cannot guarantee real-time precision in data fusion.

Although the method proposed in [8] achieves automation, it requires a large number of point-to-point data for function fitting, and the fitting accuracy is influenced by various factors, including data quality, selection of fitting functions, and data distribution. Despite there are automated camera calibration algorithms based on roadside monitoring cameras [9], [10], such methods mainly rely on vanishing point detection of vehicles and assume that the vehicle trajectory approximates a straight line or the road is straight,

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making the camera calibration accuracy susceptible to the influence of vehicle trajectories. In addition to the traditional auto camera calibration methods, researchers have proposed numerous deep learning-based camera calibration methods in recent years [11]–[14]. These methods are trained on large-scale public datasets and offer advantages such as being unaffected by traffic flow and not requiring any manual input or prior scene information. They learn semantic clues from the scene in an end-to-end manner and infer camera parameters. However, we observed that these models can exhibit noticeable prediction errors when applied in certain scenarios. Furthermore, calibration methods that rely solely on the geometric or semantic cues within the camera's field of view overlook an inherent feedback loop that ensures the camera has been accurately calibrated for the specific downstream tasks.

Apart from the camera calibration problem, another overlooked issue in spatial alignment of roadside multi-sensor systems is that existing methods mainly focus on calibrating the "radar-camera" (or "LiDAR-camera") pair. However, in practical roadside multi-camera deployment scenarios, the roadside perception system may require simultaneous calibration of a group of cameras and radars (or LiDARs). Using existing methods directly would require modifying or adjusting the system for different combinations, and the calibration process can also lead to cumulative errors, this problem becomes particularly evident in large-scale deployment scenarios.

In this paper, we investigate the challenges associated with spatial alignment in multi-view, multi-sensor deployments at the roadside, present a novel framework that utilizes geolocation cues to achieve spatial alignment. This research produced the following major contributions:

- 1) A framework for one-shot multi-view, multi-sensor spatial alignment using limited LiDAR geolocation

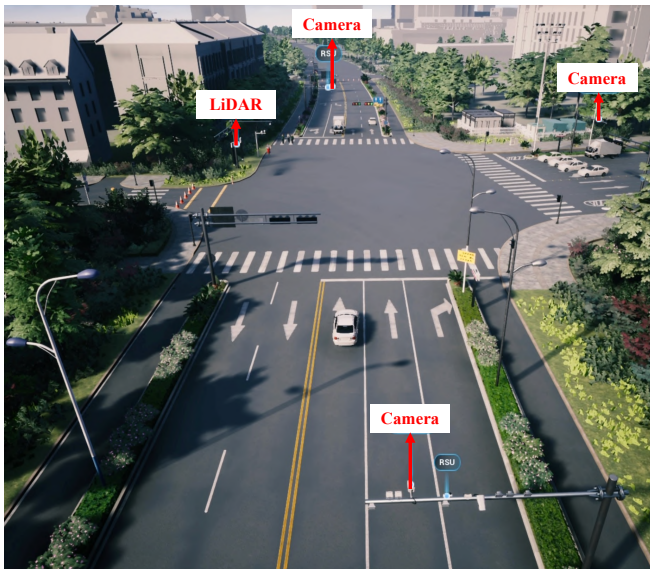


Fig. 1. Study scenario.

cues.

- 2) A monocular localization method based on deep learning, angle estimation, and distance estimation is proposed for roadside monocular scenarios.
- 3) A novel approach for approximating pseudo camera parameters is proposed to tolerate camera parameters estimation errors, position estimation errors and so on to ensure high spatial alignment accuracy.
- 4) Comprehensive experiments and evaluations were conducted in real-world scenarios to assess the framework's accuracy under different parameter settings and data requirements.

II. STUDY SCENARIO AND FRAMEWORK

The application scenario targeted in this article is depicted in Fig. 1. It includes LiDAR and multiple cameras dispersed along the roadside. We assume that LiDARs or radars have built-in object detection functions and can accurately provide the geolocations of vehicles within their monitored area. This is consistent with practical application scenarios where traffic monitoring LiDARs or radars placed along the roadside are typically calibrated. This article will use the geolocation information of vehicle targets provided by roadside LiDARs or radars to achieve spatial alignment across multiple sensors. We also assume that the roadside sensors share the same timestamp. This can be easily achieved through Network Time Protocol (NTP) time synchronization for networked roadside traffic surveillance systems. In this study, for roadside traffic surveillance cameras, which are typically used for extracting environmental information and are not meticulously calibrated during installation, a looser assumption is made. We assume that only the approximate installation position and orientation of the cameras are known, which can be readily obtained from HD maps.

The proposed spatial coordinate alignment framework, illustrated in Fig. 2, consists of multiple camera sensors and

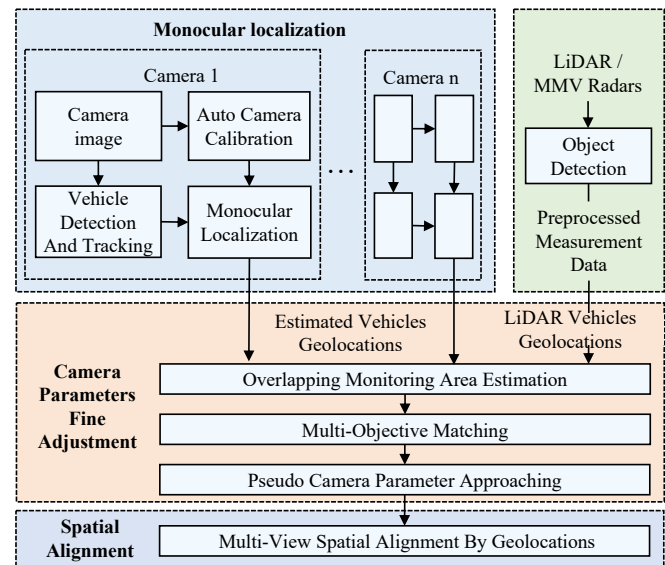


Fig. 2. The architecture of the framework.

a calibrated LiDAR or radar. We do not specify the type of LiDAR or radar used because their application interfaces are generally similar. Both LiDAR and MMV radar can provide geolocations of various vehicles within the monitored area. Furthermore, we integrate data from multiple radar units into a singular radar system. This consolidation is driven by the fact that properly calibrated multiple radars can theoretically share a unified spatial coordinate system that aligns precisely with the GNSS coordinate system.

The objective of our ongoing research is to establish spatial alignment between multiple uncalibrated, distributed camera sensors and the radar system, utilizing the GNSS framework. The framework comprises three core components: monocular localization, camera parameters fine adjustment, and spatial alignment. The monocular localization module relies solely on camera images and utilizes a deep learning-based camera calibration algorithm to estimate the initial intrinsic and extrinsic camera parameters, integrating target object distance estimation and angle estimation, enabling the estimation of geolocations for vehicles. Subsequently, the camera parameter fine adjustment module optimizes the localization results from multiple cameras by leveraging only a small amount of LiDAR data, obtaining pseudo-camera parameters. Finally, the spatial alignment module conducts a unified spatial coordinate transformation from the multi-view cameras to GNSS coordinates based pseudo-camera parameters projection matrix of different cameras. In the subsequent sections, each of these components will be introduced systematically in a step-by-step manner.

III. MONOCULAR LOCALIZATION

A. Camera Calibration

For the geometric camera model, the relationship between a 3D point $\mathbf{p}_w$ in the scene and the image pixel $\mathbf{p}_{im}$ can be expressed as:

$$\mathbf{p}_{im} = [\lambda u \ \lambda v \ \lambda]^T = \mathbf{K}[\mathbf{R} \mid \mathbf{t}][\mathbf{p}_w \mid 1]^T, \quad (1)$$

where $\mathbf{K}$ represents the camera's intrinsic parameters, and $\mathbf{R}$ and $\mathbf{t}$ denote the rotation and translation of the camera in the world coordinate system, respectively (extrinsic parameters).

This paper adopts the same assumptions as previous most studies [12], [15], [16], a pinhole camera with principal point in the image center, the camera has no skew and equal scaling in both horizontal and vertical directions. The focal length is equal in both directions, denoted as $f_x = f_y = f$, and the projection matrix $\mathbf{K}$ can be represented as $\mathbf{K} = \text{diag}([f \ f \ 1])$, where f represents the focal length in pixels. Since the range of the focal length (in pixels) is infinite and it changes when we adjust the image size, we estimate the vertical field of view (vfov) α in radians and convert it to focal length using the following equation, where h represents the image pixel height:

$$f = \frac{h}{2 \cdot \tan(\frac{\alpha}{2})} \quad (2)$$

A rotation matrix $\mathbf{R}$ can be described using three angles, which can be parameterized as pitch θ , roll ψ , and yaw

φ angles, since there is no natural frame of reference to estimate yaw (left vs right) from arbitrary images, we do not estimate yaw in our case. Therefore, we constrain the rotation to only pitch and roll components, simplifying the extrinsic rotation matrix to $\mathbf{R} = \mathbf{R}_z(\psi)\mathbf{R}_x(\theta)$. We can use the horizon line as an intuitive representation for these angles, θ is indicated by the intersection of the horizontal midpoint and the vertical center line of the image, ψ is determined by the rotation angle of the reference line in relation to the horizontal line.

Following [12], [16], We discretize the pitch θ , roll ψ , and vfov α angles into 256 bins in the spatial domain, converting the regression problem into a B-way classification problem. We utilize ResNet-50 [17] as the backbone network and employ separate fully connected layers to predict the pitch angle θ , roll angle ψ , and vfov α . The center values of each of the bins can be denoted as $\boldsymbol{\theta} = [\theta_1, \dots, \theta_i, \dots, \theta_B]$, $\boldsymbol{\psi} = [\psi_1, \dots, \psi_i, \dots, \psi_B]$, and $\boldsymbol{\alpha} = [\alpha_1, \dots, \alpha_i, \dots, \alpha_B]$, and let $\mathbf{p}^\theta = [p_1^\theta, \dots, p_i^\theta, \dots, p_B^\theta]$, $\mathbf{p}^\psi = [p_1^\psi, \dots, p_i^\psi, \dots, p_B^\psi]$, and $\mathbf{p}^\alpha = [p_1^\alpha, \dots, p_i^\alpha, \dots, p_B^\alpha]$ represent the probability mass from the fully-connected layers of each head respectively. Next, the expected value of the probability mass is calculated:

$$\begin{aligned} \hat{\theta} &= \sum_i p_i^\theta \theta_i, \\ \hat{\psi} &= \sum_i p_i^\psi \psi_i, \text{ and} \\ \hat{\alpha} &= \sum_i p_i^\alpha \alpha_i, \end{aligned} \quad (3)$$

the difference between the predicted values and the ground truth is measured using the $\mathcal{L}_2$ loss. These operations effectively address the challenges associated with regression in a continuous space.

As pointed out by [16], predicting larger focal lengths compared to the ground truth (or equivalent to smaller vfov) yields better calibration results than predicting smaller focal lengths (larger vfov). Therefore, for pitch θ and roll ψ angles, we utilize the standard $\mathcal{L}_2$ loss between the prediction and the ground truth. For the vfov α , we incorporate the Geman-McClure function [18], applying a reduced penalty when the predicted vfov is smaller than the ground truth.

$$\mathcal{L}(\hat{\alpha}) = \begin{cases} \frac{(\hat{\alpha} - \alpha)^2}{(\hat{\alpha} - \alpha)^2 + 1}, & \text{if } \hat{\alpha} \leq \alpha \\ (\hat{\alpha} - \alpha)^2, & \text{otherwise.} \end{cases} \quad (4)$$

Our total loss is defined as follows:

$$\mathcal{L}(\hat{\theta}, \hat{\psi}, \hat{\alpha}) = (\hat{\theta} - \theta)^2 + (\hat{\psi} - \psi)^2 + \mathcal{L}(\hat{\alpha}). \quad (5)$$

B. Vehicle Detection And Tracking

In this study, YOLOX [19] and Bytetrack [20] are employed for vehicle detection and tracking in surveillance cameras. Vehicle detection is performed in videos, while tracking and localization can be conducted in both video and high-definition map (HD map). Since our framework primarily focuses on intelligent vehicles, we exclusively detect vehicles and utilize the pixel coordinates of their 2D

bounding box's bottom center point as inputs for subsequent geolocation estimation.

It should be noted that using the bottom center point of the 2D bounding box as an input for the localization algorithm introduces a certain margin of error. However, experimental findings show that this approach achieves sufficient localization accuracy for high-level objective matching and data fusion. Additionally, in specific traffic deployment scenarios, if accurate associations are established among multiple sensors, roadside cameras can leverage the 3D information from LiDAR. Therefore, we did not estimate the 3D pixel positions of the vehicles in the image as inputs for localization.

C. Geolocation Estimation

After calibration, we obtain the pitch θ , roll ψ , and vfov α of the camera. In traffic scenes, the roll ψ angle deviation is generally small compared to pitch θ angle, as a result, the roll ψ angle has a minor impact on the final localization compared to the pitch θ angle. Therefore, this study does not consider the effect of the roll angle.

As shown in the Fig. 3, given the position of the camera, we only need to know the distance D and angle ω_d between the camera and the vehicle to estimate the geolocation of the target vehicle. In the context of this paper, ω_d represents the clockwise angle between the distance D and the north (N), ω_c represents the angle between the vehicle and the camera heading angle ω_h . Given an image width w and camera $hfov$, the coordinates of the bottom center of the vehicle detection box are (x, y) , ω_c can be expressed by:

$$\omega_c = \frac{hfov}{w} \left(x - \frac{w}{2} \right). \quad (6)$$

There are two different scenarios as shown in the Fig. 4 (a) and Fig. 4 (b), where the vehicle's travel direction is located on the right and left side of the camera's field of view, respectively. In both cases, (7) remains accurate [21].

$$\omega_d = \omega_h + \omega_c \quad (7)$$

The distance estimation model is depicted in Fig. 5. Assume a detected vehicle in the road scene at an (unknown) position (X_w, Y_w, Z_w) , let θ_v be the angle of a projection ray (for the camera) pointing to the intersection of the planar

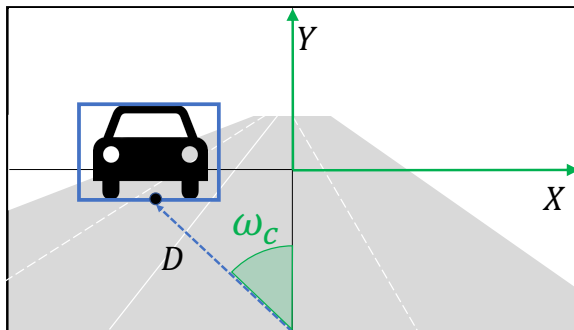


Fig. 3. The image plane used in angle estimation.

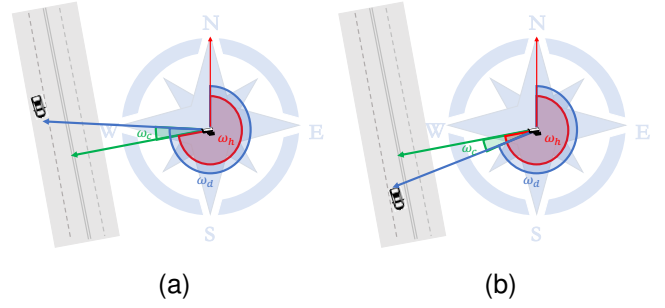


Fig. 4. The mathematical relations between ω_h , ω_d , and ω_c , along with a comparison of (a) vehicle position on the right side and (b) vehicle position on the left side within the camera's field of view.

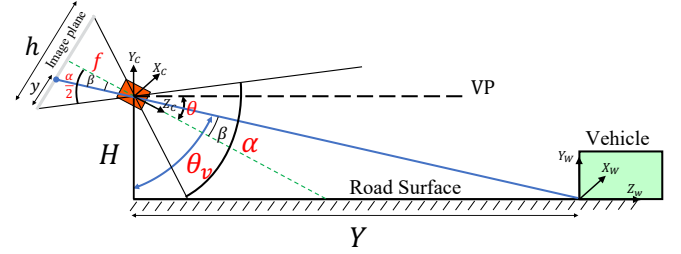


Fig. 5. Real-world vehicle distance estimation based on pixel distance information in 2D image plane.

approximation of the detected vehicle's rear or front portion with the planar road surface, H is the mounting height of the camera [22]. The longitudinal distance Y between the camera and the vehicle can be calculated by:

$$Y = H \cdot \left[\tan \left(\frac{\pi}{2} - \theta + \tan^{-1} \left(\frac{\frac{h}{2} - y}{f} \right) \right) \right]. \quad (8)$$

The distance D is given by:

$$D = \frac{Y}{\cos(\omega_c)}. \quad (9)$$

IV. CAMERA PARAMETERS FINE ADJUSTMENT AND SPATIAL ALIGNMENT

Most roadside monocular localization methods [23], [24] rely on manually selecting at least four corresponding points between the image and the map, solving for the homography matrix, and achieving the conversion from pixel coordinates to world coordinates. In practical scenarios, localization errors may be introduced due to manual point selection. In the case of the non-manual point selection method proposed in this paper, the direct transfer of camera calibration models trained on publicly available datasets to roadside scenarios, along with the utilization of predicted camera parameters for localization calculations, still struggles to meet the requirements of spatial alignment. Accurate global parameters of the camera itself are also required, including latitude, longitude, height H , and heading angle ω_h . Obtaining precise values for these parameters in large-scale traffic monitoring scenarios is extremely challenging.

To address this practical application challenge, our approach is based on treating these parameters as a combined entity. By leveraging LiDAR localization data, we optimize the visual localization accuracy while simultaneously refining this combination of camera parameters. Although some of the optimized parameters may not be the optimal or true values, the entire combination can achieve the highest localization accuracy in spatial alignment tasks, while tolerating errors in camera global parameters, improving fault tolerance and convenience, and obtaining higher robustness and accuracy in practical traffic scenarios.

A. Camera Parameters Fine Adjustment

1) *Overlapping Monitoring Area Estimation*: When optimizing camera positioning results using LiDAR localization results, the first step is to establish a correlation between the vehicle's pixel position in the image and its actual position in LiDAR, building a dataset of the vehicle's pixel changes over time and real-world latitude and longitude changes. Considering various factors such as the accuracy of the object detector, variations in the placement of the camera and LiDAR, and the varying detection ranges of each sensor, it becomes essential to filter out LiDAR data that falls outside the camera's field of view. Due to variations in latitude and longitude coordinates of each detected vehicle by the LiDAR, the corresponding angle ω_d can be computed relative to the camera's coordinates. In this study, we impose a constraint $\omega_d \in \omega_h \pm \Omega$ to selectively retain vehicles positioned in front of the camera, as depicted in Fig. 6. It is important to set Ω as small as possible to minimize the inclusion of vehicles outside the camera's field of view.

2) *Multi-Objective Matching*: Following the estimation of the monitoring area, the data association is initialized using the Kuhn-Munkres algorithm [25], [26]. The correct data association is a prerequisite to ensure the convergence of the loss during the optimization of camera parameters,

and the primary objective of this step is to generate a small dataset for subsequent training purposes. Typically, the data association is carried out once, assuming reliable associations. Nevertheless, in cases where the optimization of camera parameters faces difficulties in loss convergence due to incorrect associations, the association is performed repeatedly until the loss converges successfully, ensuring the accuracy of the camera parameter optimization. Even if we slightly underestimate Ω and filter out some vehicles that are originally within the camera's field of view, it still serves the purpose to match a reduced set of vehicles, as demonstrated in the subsequent experiments in this paper.

3) *Pseudo Camera Parameter Approaching*: As the localization algorithm is differentiable, the losses between the predicted distances and angles from the model and the ground truth obtained from the LiDAR can be calculated. By utilizing backpropagation [27] and stochastic gradient descent, the camera parameters can be continuously updated to enhance the localization accuracy. As in (2), there exists a deterministic relationship between the focal f and the vfov α . Therefore, during the optimization process, we only compute gradients for the α , which is then used to update the values of the angles f and $hfov$. Let's assume the predicted angle error is $\hat{a}_i$ with the true value a_i , and the predicted distance estimation error is $\hat{d}_i$ with the true value d_i . Given n samples, geolocation loss $\mathcal{L}_g$ is determined by the following formula:

$$\mathcal{L}_g = \frac{1}{n} \sum_{i=1}^n \left((\hat{a}_i - a_i)^2 + (\hat{d}_i - d_i)^2 \right). \quad (10)$$

We minimize using stochastic gradient descent with the Adam optimizer [28], employing an initial learning rate of $\eta = 0.001$. The algorithm's simplicity in operations contributes to its fast training speed and high tolerance towards the amount of data, as evidenced by the experimental results.

B. Multi-View Spatial Alignment According to Geolocations

When the parameter combination for the i -th camera, denoted as A_i , is obtained, and the pixel position of a certain object within the common detection area in the i -th camera is given by $p_i = (x_i, y_i)$, its corresponding geolocation q_i can be represented as $q_i = (l_i, g_i)$. If there are a total of n cameras deployed in the scene, the parameter combination for the multi-view cameras system is denoted as $(A_1, A_2, \dots, A_n)$. The overall process of multi-view spatial alignment can then be expressed as follows:

$$(A_1, A_2, \dots, A_n) \begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix} = \begin{pmatrix} q_1 \\ q_2 \\ \vdots \\ q_n \end{pmatrix}. \quad (11)$$

Using the above method, we can simultaneously solve for the parameter combinations of all cameras in the scene.

V. EXPERIMENTAL RESULTS

Our experiment data is collected from two intersections and one highway in Hangzhou, China.

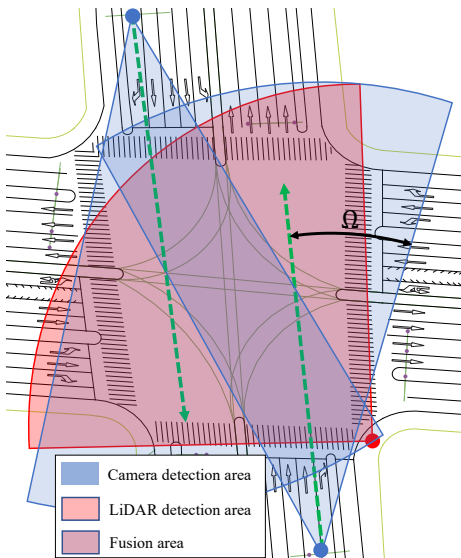


Fig. 6. The fusion area of the cameras and the LiDAR.

A. Localization Evaluation

The Pano360 dataset [16] was used for training camera calibration model. Fig. 7 reveals the impact of different traffic flows on the accuracy of camera calibration. It was observed that our camera calibration model is affected to some extent by the presence of traffic, but the resulting errors remained within an acceptable range.

We evaluated the localization algorithm in two scenarios within a single intersection: straight driving and turning conditions as show in Fig. 8(a) and Fig. 8(b). All vehicles from two scenarios were combined into a cohesive dataset consisted of only 450 samples from 12 vehicles, which, in real-world scenarios, would require only 4 seconds of data collection. Fig. 9(a) and Fig. 9(b) present a comparison of the localization results before and after optimizing camera parameters. Our observations indicate a significant improvement in localization accuracy achieved by the optimization algorithm, particularly for vehicles located at greater distances from the camera. We conducted a statistical analysis of distance errors in the dataset, both overall and for three individual vehicles in Fig. 11, without optimizing camera parameters. The distribution of distance absolute error, as depicted in Fig. 10(a), showed that the majority of errors were within a range of 2.5 m. Variations in errors were observed at the same distance, mainly due to differences



Fig. 7. The impact of different traffic flows on the accuracy of camera calibration. The green and red lines represent the true and predicted horizontal lines respectively.



Fig. 8. Scenarios at Intersection 1: (a) Straight Driving, (b) Turning.

TABLE I
SELECTED PARAMETERS AND RMSE VALUES

Selected Param	RMSE-D (m)	RMSE-A (°)
before optimization	9.24	1.57
f	2.54	0.71
H	7.03	1.57
θ	5.18	1.57
$f + \theta$	2.00	0.66
$f + H$	2.41	0.66
$H + \theta$	1.15	1.57
$f + H + \theta$	1.11	0.67

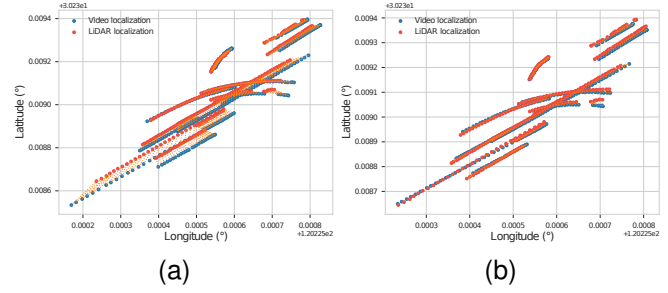


Fig. 9. Analyzing Localization Results through Vehicle Trajectory Overlays in Three Scenarios: (a) Before Camera Parameter Optimization, (b) After Camera Parameter Optimization

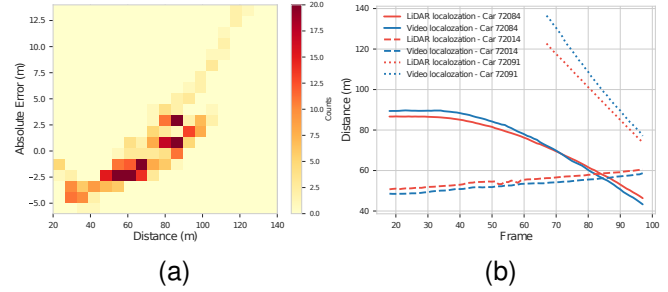


Fig. 10. Distance Error in Monocular Localization: (a) Heatmap of Distance Error Distribution, (b) Frame-wise Analysis for Three Vehicles.

in angle estimation errors. Additionally, as shown in Fig. 10(b), the trajectories of the three vehicles were well-fitted, with relatively larger errors observed as the actual distance increased.

B. Approximating Evaluation

During the process of optimizing localization accuracy, we conducted comparative experiments on different parameter combinations where errors were introduced to the original parameters to enhance the comparison results. The results from TABLE I show that our algorithm is most affected by the camera focal length f , followed by pitch θ , and least affected by height H , as indicated by the distance root mean square error (RMSE-D) and angle root mean square error (RMSE-A). Incorporating the approximation of the focal length reduces the angle error and improves distance accuracy for pitch θ and height H . Optimizing all three parameters together results in the highest accuracy for the localization algorithm.

In addition to evaluating the impact of camera parameters, we investigated the influence of dataset distribution on the optimization algorithm by jointly optimizing the focal length f , pitch θ , and height H , as presented in Fig. 11 and TABLE II. We observed that data distribution within specific distance ranges influenced the accuracy of the approximation. Testing with vehicles covering longer distance ranges led to decreased distance and angle errors. By selecting only straight-driving vehicles for optimization, the method effectively reduces angle errors, as it simultaneously optimizes both the distance and focal length f , leading to angle optimization as

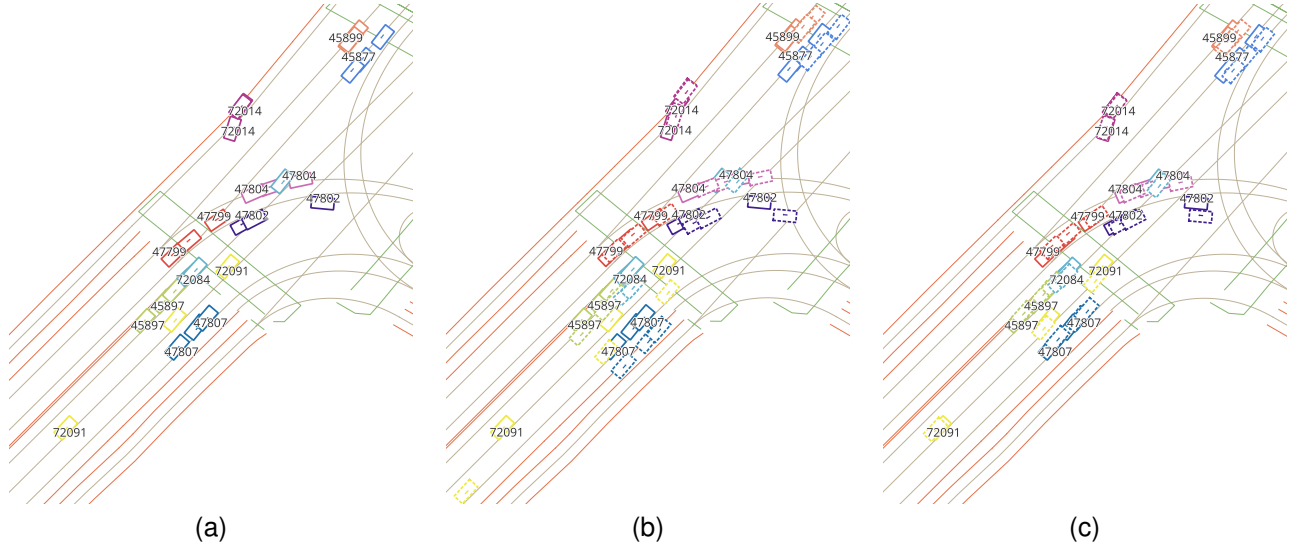


Fig. 11. Localization Results in Intersection 1: (a) Ground Truth, (b) Before Camera Parameter Optimization, (c) After Camera Parameter Optimization.

TABLE II
SELECTED DATA AND RMSE VALUES

Selected Data	RMSE-D (m)	RMSE-A (°)
before optimization	9.24	1.57
45899	37.11	1.73
72084	1.47	0.77
45899 + 72084 + 72091	1.38	0.72
47799	1.82	1.05
47804	4.13	0.72
47799 + 47804	1.76	0.73
Total Data	1.11	0.67

than the angle error. For spatial alignment purposes, a 2 m camera position error range is deemed acceptable, while for pure localization, a wider acceptable error range can be considered.

C. Alignment Evaluation

To evaluate the applicability of our framework in a multi-camera scenario, we selected an intersection scene monitored by two cameras, as shown in the Fig. 13. The majority of vehicles were detected simultaneously by both cameras, while a few vehicles were not detected and localized by the other camera due to the camera's viewing angle. The localization results of the two cameras were compared with the LiDAR localization on the HD map. It can be observed that the localization results of both cameras closely match the results from the LiDAR localization, demonstrating the effectiveness of our framework in multi-camera scenarios.

VI. CONCLUDING REMARKS AND FUTURE WORK

In this paper, we proposed a framework for spatial alignment in multi-view cameras and LiDAR scenarios based on geolocation cues. Real-world experiments demonstrated that our monocular localization algorithm, after LiDAR-based optimization, achieved a distance error of only 1.1 m, making it suitable for spatial alignment. This framework eliminated the need for individual alignment of sensor pairs and enabled direct spatial alignment of sensor groups, significantly reducing the manual costs and errors associated with spatial alignment, maximizing the utilization of multi-sensor data. It offered advantages of low maintenance costs, easy scalability, and robustness in various road and environmental conditions and could also be applied in MMV radar scenarios.

Our framework can also integrate various visual tasks, including vehicle pose estimation, semantic segmentation, and velocity estimation; by leveraging HD map, it enables intelligent roadside decision-making and path planning;

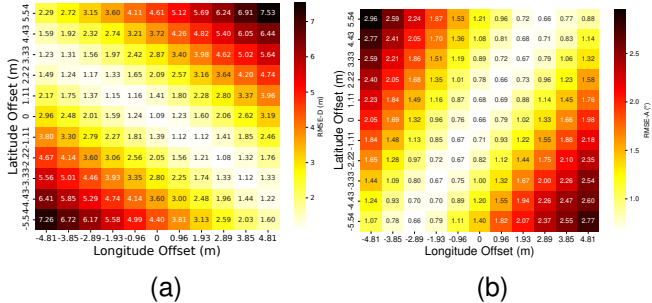


Fig. 12. Impact of Latitude and Longitude Errors of Camera on Distance and Angle Estimation: (a) RMSE-D Result, (b) RMSE-A Result.

well. Additionally, even with turning vehicles, the optimization results from straight-driving vehicles sufficed to achieve accurate approximation.

Considering the camera's latitude and longitude errors during deployment, we simulated different error scenarios at intersection 1 and observed an increase in distance and angle estimation errors as errors grew as shown in Fig. 12(a) and Fig. 12(b). The trends differed significantly, with angle errors deviating less in the heading angle direction, and distance errors deviating less in the direction perpendicular to the heading angle. Overall, the distance error is larger

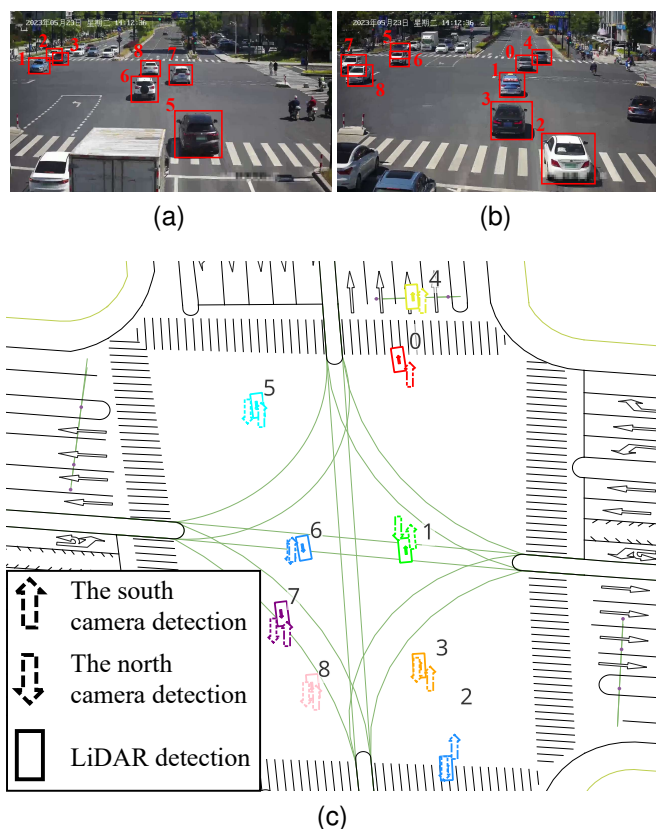


Fig. 13. Alignment Evaluation: (a) The North Camera Image, (b) The South Camera Image, (c) The Spatial Alignment Result.

through data association with multi-sensor, our framework enables the creation of a comprehensive roadside perception dataset by combining information from multiple sensors. Therefore, future research will focus on exploring fusion strategies and algorithms to integrate diverse data sources and improve roadside intelligence.

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